Yliopistokatu 4 A 308, OULU 90570

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tharindu326@gmail.com

SUMMARY

Reliable and ambitious technology enthusiast with innovative thinking skills seeking further exposure in AI and a challenging opportunity to pursue a career in the domains of deep learning, machine learning and computer vision

SKILLS

- **Programming Languages**: Python
- **ML Tools**: PyTorch, TensorFlow, Keras, Numpy, Matplotlib, scikit-learn, OpenCV
- **Object Detection/Tracking**: YOLOv4 v5 v8, Mask-RCNN, Faster-RCNN / DEEP-SORT, ByteTrack, StrongSort, BotSort
- **OCR:** TesseractOCR, PaddleOCR
- **Facial Recognition**: FaceNet, VGGFace, Dlib, OpenFace, DeepFace, Arcface
- **Image/Video Classification**: ViT, InceptionResnetV2, VGGnet, Inception, Xception, ResNeXt, CVNets.
- **Image Segmentation**: Mask-RCNN, U-NET, YOLOv8 v7
- **Pose Estimation**: OpenPose, VitPose
- **NLP**: Image Captioning (LSTMs, CNNs), LLMs (LLAMA, GPT, RAG systems)
- **GAN:** DC-GAN
- **Regression and Classification**: SVM, Decision Trees, Naive Bayes
- **Development/Operations Tools**: GitHub, Flask, Docker, Databases (MySQL, MongoDB), Linux, AWS (S3, EC2, RDS), GCP
- Data Mining
- **Edge Devices**: Nvidia Jetson (Nano, Orin, AGX Xavier), Raspberry pi , Coral Edge TPU accelerator, Hailo AI processor
- **Deployment tools**: Flask APIs, AWS Sagemaker
- Teamwork and Collaboration
- Project Management
- Client Relationships
- Research and analysis

Tharindu Ekanayake

EXPERIENCE

October 2024 - Current
Research Assistant Center for Machine Vision and Signal Analysis, University of Oulu | Oulu

- I am currently contributing to the AugmentCare project, which aims to enhance understanding of human physiological and behavioral states within virtual reality (VR) environments. Leveraging advanced technologies such as computer vision, signal processing, and machine learning, our research focuses on extracting motion and physiological signals from the human body. We employ techniques like segmentation and pose estimation to identify patterns that reveal insights into cognitive load and stress responses. Both supervised and unsupervised machine learning approaches are being applied to analyze these features, enabling accurate classification of user states and the development of adaptive, personalized VR experiences based on real-time physiological feedback.

June 2024 - September 2024
Internship Machine Vision and Signal Analysis Center, University of Oulu | Oulu

- My responsibilities encompassed planning the AI at the Edge Bachelor's Thesis Project Course. I was organizing project topics, creating materials, and handling technical aspects like configuring and setting up devices. I trained and converted models using TensorFlow and PyTorch for object detection and classification, converted TensorFlow models to TensorFlow Lite with quantization for Coral TPU accelerators, and benchmarked model performance. Additionally, I planned thesis topics and prepared demonstrations to showcase AI applications, using these insights to develop detailed project content and scope.

July 2021 - December 2023
Machine Learning Engineer ZOOMi Softlab | Colombo, Sri Lanka

- Design and Engineer Machine Learning Systems / Investigate and Apply Machine Learning Techniques and Instruments / Create Artificial Intelligence Solutions and Algorithms for Automation Objectives.
- Participated in projects involving AI technologies such as object detection, segmentation, classification, object tracking, and Large Language Models with Retrieval-Augmented Generation (RAG) systems.
- Developed LayerNext, a platform for data management and training of computer vision models.

June 2021 - July 2023
Computer Vision Engineer (On Contract) nSight Surgical | San Francisco, CA

- Developed a Machine Learning framework focused on enhancing patient safety in hospitals and optimizing surgical workflows / Utilized object detection models (YOLO), pose estimation tools (VitPose, OpenPose, YOLOv7), object tracking algorithms (DeepSORT, ByteTrack, OcSORT, StrongSORT), and inference systems (OpenCV, Pytorch, ONNX). Implemented multi-processing and communication pipelines (Web-Sockets, ZMQ) along with REST-APIs.

October 2020 - July 2021
Engineer - Converged Charging Dialog Axiata PLC

- Development and operations in converged charging system (OCS) / project planning / provisioning, and deployments for charging network.

EDUCATION

May 2025
Master of Science | Computer Science and Engineering, AI
University of Oulu, Oulu, Finland

April 2019
Bachelor of Science | Electrical and Electronic Engineering
University of Peradeniya, Sri Lanka, Kandy

REFERENCES

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